General comments on EE2-21 Feedback Systems paper 2013

1. The students have done very well on this question, scoring approximately 77%.
   (a) This is an electric circuit modeling question and is a somewhat typical study group question.
      i. Typical study group question.
      ii. Typical study group question.
      iii. Typical study group question, asks for a physical interpretation.
      iv. A bit tricky since it uses all the results above.

   (b) This is a Nyquist diagram/Routh-Hurwitz question and is mostly typical of study group questions.
      i. Typical study group question.
      ii. Typical study group question.
      iii. Typical study group question.
      iv. Typical study group question.

2. This question combines knowledge about Nyquist analysis and the Routh-Hurwitz criterion in a slightly non-standard way for compensator design. The students did less well on this question, scoring an average mark of 55%.
   (a) Standard study group question, although part (iii) needs a little thought to deduce the phase margin.

   (b) This uses the extended Nyquist stability criterion in that it requires the determination of closed-loop stability for all possible gains. The students tend to make elementary mistakes in signs, inversions and inequalities.

   (c) Standard study group question.

   (d) Standard study group question, although they haven't done PD controllers in that context.

3. This is a PD control design question that uses basic concepts of stability from a system's characteristic equation. The students did relatively well scoring approximately 61%.
   (a) Most students got the characteristic equation right, but some did not give the correct comment on the closed-loop stability.

   (b) Most students got the location of closed-loop poles right.

   (c) Most students seem to have understood the concept of PD compensator design.

   (d) Some students found the implementation of a controller using opamps a little difficult.